

CHAPTER 1: INTRODUCTION

1.1 Introduction

FashionPriceX – Smart Dynamic Pricing System for Fashion Products is an innovative web-based e-commerce application created to improve the online clothing retail market through smart pricing. Unlike other platforms offering the same services, which sell products at a constant price, this system allows for dynamically changing prices based on demand, supply, and other relevant factors. The application also offers special dashboards for customers, sellers, delivery personnel, and managers to manage items, make transactions, operate on the territory, and ensure maximal security during the selling process. Advanced technologies coupled with the application’s use of AI concepts can improve the seller’s profit and overall customer satisfaction.

1.2 Background

FashionPriceX – Smart web-based system for dynamic pricing of fashion products is an innovative web-based e-commerce platform intended to provide online clothing trading with artificial intelligence-driven demand-aware pricing. Unlike conventional e-commerce systems, FashionPriceX allows adjusting the prices of products in accordance with demand, supply, and other relevant factors. The platform offers separate interfaces for customers, sellers, delivery agents, and administrators to ensure the reliable management of items, online shopping, sales, and delivery operations. The project is expected to integrate concepts of artificial intelligence with cutting-edge web technologies to facilitate informed price decisions, enhance customer experience, and maximize profits.

1.3 Problem Statement

Fashion e-commerce platforms traditionally offer goods at a constant price, which often fail to meet the actual demand and result in losses for sellers. At the same time, manual price adjustments are time-consuming and may prove to be ineffective in terms of aligning supply and demand. Besides, dynamic pricing mechanisms are hardly used by e-commerce platforms due to the complexity of individualizing customers’ preferences and estimating prices in real time. In addition, it is challenging for fashion sellers to identify the optimal price for maximizing profits, while customers have limited opportunities to purchase high-quality offers at an attractive price.

1.4 Project Objectives

The proposed project pursues the following goals:

- Design a fashion e-commerce platform that would maximize business profits and ensure an overall satisfaction of customers;
- Offer a dynamic pricing system that allows maximizing profits while ensuring the reliability of transactions to minimize losses;
- Implement the platform as a reliable marketplace to ensure the safety of online transactions, as well as convenient product management, shopping, and delivery operations;

1.5 Scope of the Project

FashionPriceX aims to create a reliable online marketplace for fashion clothing products with an integrated dynamic pricing system. The project focuses on implementing the system’s basic features, enabling users to register, log in, browse products, add desired items to a shopping cart, make payments securely, manage orders, and deliver products. In the future, the system will offer additional functions, including personalized product suggestions based on individual preferences, demand estimation, cloud storage, and other relevant features.

1.6 Existing System

Fashion e-commerce platforms traditionally offer products at a constant price, which rarely reflects the actual demand-supply ratio. Most platforms allow customers to use e-commerce services, including browsing products, managing a shopping cart, and making payments. However, most conventional e-commerce systems offer limited functionality in terms of dynamic pricing and thus cannot estimate the actual demand for products and respond to sudden changes in supply-demand ratios. As a result, platforms are unable to offer competitive pricing, which leads to losses for sellers due to low sales volumes and the inability to attract customers.

1.7 Proposed System

FashionPriceX proposes to introduce a dynamic pricing system that estimates demand, supply, and other relevant factors to ensure that prices are adjusted automatically in accordance with current supply-demand ratios. The new system will also offer enhanced security measures, including user

authentication, to ensure that only authorized users can access the system’s features. In addition, the project will expand the set of functions offered by e-commerce platforms, including products management, shopping cart, payment system, order management, delivery operations, and others. Overall, the proposed system will utilize artificial intelligence to facilitate informed decisions, maximize profits, and ensure the security of online transactions.

1.8 Feasibility Study

It is feasible to accomplish the project due to its technical, economic, and operable feasibility. First, the system can be designed and developed with the use of Python programming language, Flask, HTML, CSS, JavaScript, Bootstrap, and MySQL. These technologies are appropriate for the project since they offer competitive advantages over conventional solutions. In addition, the project’s economic feasibility is projected due to the potential benefits of the platform, including the maximization of profits on both seller and customer sides, enhanced user experience, and adherence to the latest technological trends. In terms of operability, the platform will offer intuitive interfaces to ensure seamless operations. The system will be implemented with minimal complexity, which proves its feasibility in terms of time management and operability.

1.9 Organization of the Report

The report consists of five chapters that describe various aspects of the project. The first chapter introduces the project and provides background information on it, defines the objectives of the study, states a problem, as well as the scope, feasibility, and organization of the report. The second chapter contains a literature review and related research paper. The third chapter focuses on system analysis, design, and development, whereas the fourth chapter discusses the implementation process and results. The final chapter highlights the conclusion, limitations, future work, and references.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

The literature survey consists of the research and technologies related to dynamic pricing, e-commerce, and intelligent pricing mechanisms. It also covers the features, advantages, and limitations of the existing works related to the proposal. Additionally, it includes the justification and motivation for the proposed FashionPriceX system. The current chapter provides the background of the study, a review of the related works, comparative analysis, gap analysis, and proposal’s motivation. Therefore, it justifies the necessity of the new project based on the previous research results and peculiarities of the existing systems.

2.2 Background

Online shopping has become an essential part of people’s lives, resulting in the continuous need for new e-commerce developments. The majority of fashion e-commerce systems concentrate on product management, secure purchasing, and customer-friendly interfaces. However, most of them utilize conventional pricing strategies. The recent achievements in Artificial Intelligence (AI), Machine Learning (ML), and Reinforcement Learning (RL) areas enable organizations to apply intelligent ways for dynamic product pricing based on supply-demand, customer behavior, and other significant factors. Thus, the current developments in AI, ML, RL contribute to the creation of the smart pricing mechanisms that help companies improve their performance and satisfy customers’ needs.

2.3 Review of Existing Studies

The research works related to the development of the FashionPriceX system include the articles about intelligent pricing and smart e-commerce. They describe the possibilities of utilizing smart e-commerce systems and recommend applying Machine Learning methods for dynamic pricing. The literature review also comprises the studies exploring different strategies to ensure effective pricing mechanisms based on supply-demand, inventory, and other factors. Apart from that, the existing projects present various e-commerce systems where the majority of them offer limited functionality related to pricing or lack smart technologies.

2.4 Comparative Analysis

The majority of the existing systems offer the fundamental features of online shopping platforms, including browsing the catalog, making purchases, tracking orders, and other basic functionality. In addition, most of them apply traditional pricing methods, whereas some of them provide smart pricing mechanisms that require advanced data processing techniques. However, the current projects are not focused on the e-commerce systems for fashion retailers, and some of them are applicable to specific industries, such as airline or hotel booking. Thus, FashionPriceX system can be considered as an effective solution that offers an integrated approach to developing an e-commerce platform for fashion retailers with intelligent dynamic pricing mechanisms. The proposed project is intended for small and medium-sized fashion businesses, whereas other systems are applicable to any organization, including the companies that do not offer fashion products.

2.5 Research Gap Analysis

The literature review and analysis of the existing systems indicate that the current smart e-commerce systems offer limited functionality related to dynamic pricing and do not consider inventory, customer purchasing behavior, and other aspects of product sales. The majority of dynamic pricing mechanisms are complicated, requiring accurate data processing and advanced technologies. Moreover, most of the e-commerce projects are expensive, focusing on large organizations, and not web-based. Thus, the development of the web-based e-commerce system with an integrated approach to dynamic pricing based on inventory, supply-demand, and other factors contributes to filling the gap in smart technology application in fashion retail sales.

2.6 Motivation for the Proposed Work

The motivation for the development of the intelligent pricing system relates to the necessity to improve fashion retailers' performance and ensure customer satisfaction. The integration of the web-based e-commerce platform and dynamic pricing mechanisms will offer online sellers a possibility to analyze the data to determine optimal product pricing, increase inventory turnover, and ensure positive customer experience and satisfaction. Apart from that, the proposed project will also serve as the opportunity to apply AI and ML concepts to explore product sales peculiarities and solve the particular task related to inventory turnover in the current business environment.

2.7 Chapter Summary

To summarize, the current chapter presents the literature review and analysis of the existing studies and projects related to the smart e-commerce system. It covers the background, research work, comparative analysis, gap analysis, and motivation for the proposed FashionPriceX project. The key findings include the characteristics of the existing e-commerce platforms, the opportunities and limitations of the studies, and the necessity for the integrated approach to developing a dynamic pricing mechanism in the e-commerce system. Therefore, the chapter justifies the need for the new project based on the findings and describes its role in the smart e-commerce development.

CHAPER 3: SYSTEM REQUIREMENTS

3.1 Introduction

The System Requirements chapter entails the hardware, software, development tools, and technologies required to design and develop the FashionPriceX – Smart Dynamic Pricing System for Fashion Products successfully. In addition, it describes the application’s features that enable it to provide a secure environment for managing products, conducting transactions, and implementing dynamic pricing strategies.

3.2 Hardware Requirements

The following are the minimum hardware requirements needed to develop and operate the proposed application:

- Processor: Core i3 or better
- RAM: 8 GB RAM or better
- Storage: 256 GB storage or better
- Internet Connection: A computer with a broadband internet connection for development and testing purposes

3.3 Software Requirements

The following software tools are needed to develop the proposed application:

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- Web framework: Flask framework
- Frontend Technologies: HTML, CSS, JavaScript
- Database: MySQL
- Database Tool: MySQL Workbench
- Web Browser: Google Chrome or Microsoft Edge browser

3.4 Development Tools

The following development tools were used to build the project:

- Visual Studio Code (VS Code)
- Python IDLE
- MySQL Workbench
- Google Chrome Developer Tools

3.5 Technology Stack

The project utilizes Python and Flask framework to develop the application backend, which handles the application’s business logic layer. In addition, it uses HTML, Cascading Style Sheets (CSS), JavaScript, BootStrap, and Jinja2 to design and develop its frontend application programming interface (API). Furthermore, the project implements a MySQL database to manage and store its application data. Finally, it applies concepts in Machine Learning/Reinforcement Learning to build its dynamic pricing engine, thus enabling it to provide quality-assured products at affordable prices.

3.6 Software Architecture

FashionPriceX uses a three-tier software architecture to develop the application. It implements a Presentation tier that offers an interface for sellers, delivery personnel, customers, and administrators. It also utilizes the Application tier, which is built using the Flask framework, to execute the application’s business logic functions. These functions include user authentication, product management, dynamic pricing engine, payment system, and order management.

3.7 Summary

This chapter described the project’s system requirements by identifying the hardware and software needed to develop and operate the FashionPriceX – Smart Dynamic Pricing System for Fashion Products. In addition, it highlighted the development tools, technology stack, and software architecture used to build the project. The identified tools and technologies are efficient and reliable and will enable the project to deliver smart dynamic pricing and full-featured e-commerce platform services.

CHAPTER 4: SOFTWARE REQUIREMENT SPECIFICATION

4.1 Introduction

The SRS outlines the requirements for developing the FashionPriceX – Smart Dynamic Pricing System for Fashion Products. It also covers the testing and maintenance processes for the application. The document describes the features, interface, and behaviors of the system under development. It ensures that the software system is fit for proposed purposes and that it will satisfy all user requirements during its operation.

4.2 Overall Description

FashionPriceX is a web-based fashion e-commerce platform that offers a dynamic pricing approach and a complete set of online shopping features. The application enables customers to search for and compare fashion items, add selected products to their shopping carts, make payments, and track their orders. The sellers can add and update products, view their inventory, and track the sales of their items. The administrators manage the application by adding and removing users, organizing product categories, overseeing the entire process, and reviewing reports. Delivery personnel can update delivery information and manage their assigned orders. The system is designed based on a three-tier software architecture that supports the presentation, application, and data layers in order to ensure effective communication and security of information within the application and between the users and the database.

4.3 User Characteristics

There are four categories of users of the proposed system, namely, customers, sellers, administrators, and delivery personnel. All users have different roles, which determine their behaviors and access levels to the application’s features.

- Customer: A customer may register, log in, browse through the available products, search for desired items, add items to their shopping cart, pay for the selected products, and track their order.
- Seller: A seller can add and update products, manage their inventory, view their customers’ orders, monitor their sales, and update the prices of their items by using the dynamic pricing module.

- Administrator: An administrator oversees the operations of the application. The admin adds and deletes users, manages product categories and product data, sets pricing rules and other policies, uploads and views reports, and monitors the application’s performance.
- Delivery Personnel: A delivery agent receives assigned orders, updates the delivery status, and completes the delivery of the products to the customers.

4.4 Functional Requirements

The following key services will be provided by the FashionPriceX system under development:

- The system allows users to register and log in to the application by using secure passwords.
- The application enables customers to browse through the available products and purchase items using the provided search feature. The buyers can add their chosen items to the shopping cart, make payments, and track the delivery progress of their orders.
- The FashionPriceX system enables users to search and purchase products based on their preferred category. The application also allows sellers to update and manage their products, monitor their inventory, and view and manage their sales data.
- The dynamic pricing module updates the price of products according to the supply-demand trends of the fashion items.
- The system provides the admin with the authority to manage the application by adding users, organizing product categories, managing products, viewing reports, and monitoring the overall performance of the application. The admin can also update the prices of products.
- The delivery module assigns delivery tasks to the delivery personnel. Moreover, the delivery staff can update and track the progress of the delivery of products to the customers.

4.5 Non-functional Requirements

- The system to be developed should meet the following key quality requirements to ensure it is efficient and meets the needs of the intended users.
- The FashionPriceX application will be developed to offer enhanced performance, including fast response time, increased speed, and high scalability. The system will support multiple users and guarantee high performance without experiencing breakdowns or failures.

- The application will provide users with a user-friendly interface that is interactive and compatible with different web browsers and devices. The system will be highly secure because it will use encrypted passwords, secure log in the process, and other features to protect user data. The application will ensure effective data management, including regular data backup, to prevent the loss of valuable information. The software system will be ported on different platforms and servers, and it will remain functional without requiring modifications to its structure or programming. Additionally, it will be easy to maintain, operate, and modify because it will be modularly designed. It will also be compatible with multiple operating systems.

4.6 Use Case Diagram

A Use Case Diagram is a UML diagram which explains the interaction among the user(s) and the FashionPriceX system. It also explains the significant functions performed by the system and how different users interact with the system. The main actors in the FashionPriceX system are Customer, Seller, Administrator, and Delivery Personnel. Customers have various functions such as registering and logging in to the application, searching for and browsing products, adding products to their cart, placing an order, making payments, and tracking the delivery of the products. In addition, sellers can add, update, edit and delete fashion products, and manage the inventory products. Moreover, the administrator monitors the whole process by managing the users, products, categories, reports, and pricing. In addition, Delivery Personnel updates the delivery information of an order placed by the customers. Therefore, the Use Case Diagram below is an integral part of the developed application as it highlights the functions performed by the system and how these functions support different users who interact with the system.

4.7 Use Case Description

The major use cases of the Fashion PriceX include:

- **User Registration:** This use case applies when a new user wants to register with the system by providing their details such as name, email address, mobile number, and password.
- **User Login:** This use case applies when a user logs in with a given user name and password to get access to the system functions.
- **Product Management:** This use case enables the seller to add, update, edit, and delete fashion products, pictures, descriptions, prices, and inventory.

- **Order Processing:** This use case enables the customers to order the selected items and make payments to clear the purchase amount. In addition, this use case also enables the seller and administrator to monitor the progress of the ordered items.
- **Payment Processing:** This use case enables the system to process the mode of payment selected by the customer such as online and cash on delivery, while also recording the payment details in the system.
- **Order Tracking:** This use case enables customers to track the status of the ordered items, whereas the delivery personnel updates the delivery status of the items up to completion of the delivery process.
- **Administration:** The administration use case enables the administrator to perform a range of functions such as managing the users, products, categories, reports, and policies related to pricing of products.

4.8 External Interface Requirements

FashionPriceX application has different interfaces to interact with different users. The User Interface for the FashionPriceX will involve the use of HTML, CSS, JavaScript, Bootstrap, and Jinja2. The Hardware Interface involves personal computers, laptops, smartphones, and internet connections. The Software Interface will involve Python, Flask, MySQL, MySQL workbench, and Google Chrome among others. In addition, the Database Interface will involve connecting the application to the database via application servers to ensure that all data about users, products, inventory, payments, and orders are captured and stored appropriately in the system. The system will also have an interface with online payment platforms to ensure that payments are processed digitally using online mode of payments.

4.9 Constraints

Some of the constraints that can be expected for the FashionPriceX application include; the need for internet connections to access the application and make online payments. In addition, there is a need to make sure that the system has adequate information regarding inventory and sales trends to be able to update prices dynamically. The users also need to provide their personal details to access the application. The application will require users to operate it using web browsers such as Google Chrome and Microsoft Edge among others.

4.10 Assumptions

Among the many assumptions made in the development of this project, some of the key assumptions made include the following: it is assumed that all users are computer literate and can therefore operate the application successfully. In addition, it is assumed that sellers provide the required information about the products such as inventory details and update them regularly. In addition, it is also assumed that the customers provide true information about their personal details when registering and also pay their amounts due to the company upon receiving the items purchased. It is also assumed that there are no constraints with regard to internet connections, servers, and databases that can hinder proper functioning of the system. Lastly, it is assumed that the project can be extended to include such features as making recommendations based on demand, forecasting demand based on inventory data, and deploying the application to the cloud.

4.11 Summary

This chapter introduced Software Requirement Specification of the FashionPriceX system by discussing the overall functions, user characteristics, functional and non-functional requirements, use case description and diagrams, external interface requirements, constraints, and assumptions about the project. The discussion helped reveal the expectations of the system by defining the functions that it will perform, the quality standards expected, and the extent to which it will be used. The Software Requirement Specification document thus plays a critical role in supporting the development processes of new products and services in accordance with the intended performance objectives.

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

The System Design Chapter entails the design of the FashionPriceX – Smart Dynamic Pricing System for Fashion Products, including the system’s architectural design, data flow diagrams, database design, and user interface, as well as UML Diagrams for illustrating the system components.

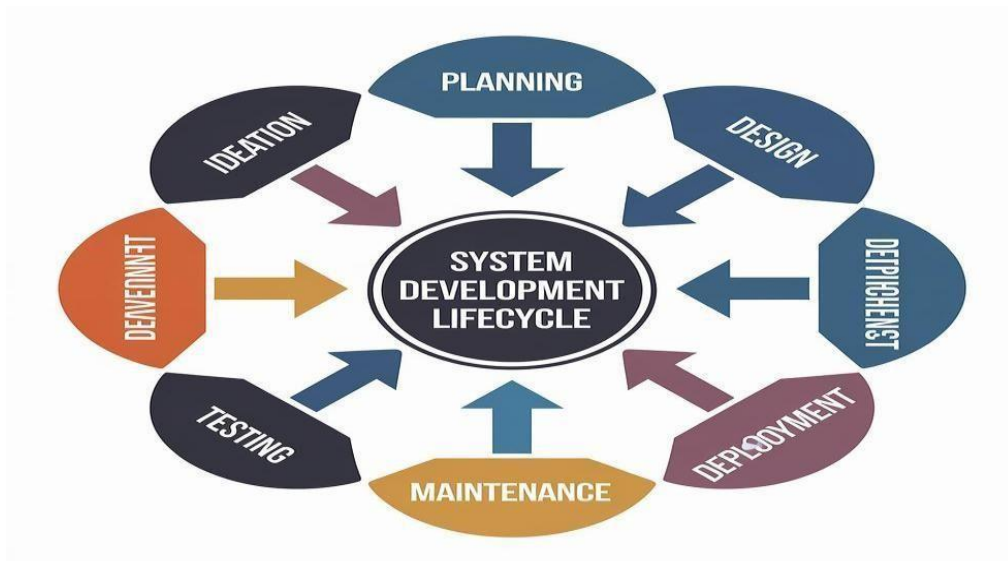


Fig: 5.1.1: System Design

5.2 System Architecture

The FashionPriceX employs a three-tier system architecture in which the presentation layer is designed using HTML, the application layer is designed using Flask to manage the business logic, authentication, dynamic pricing, orders, and payments, and the database layer is designed using MySQL to store the system’s data.

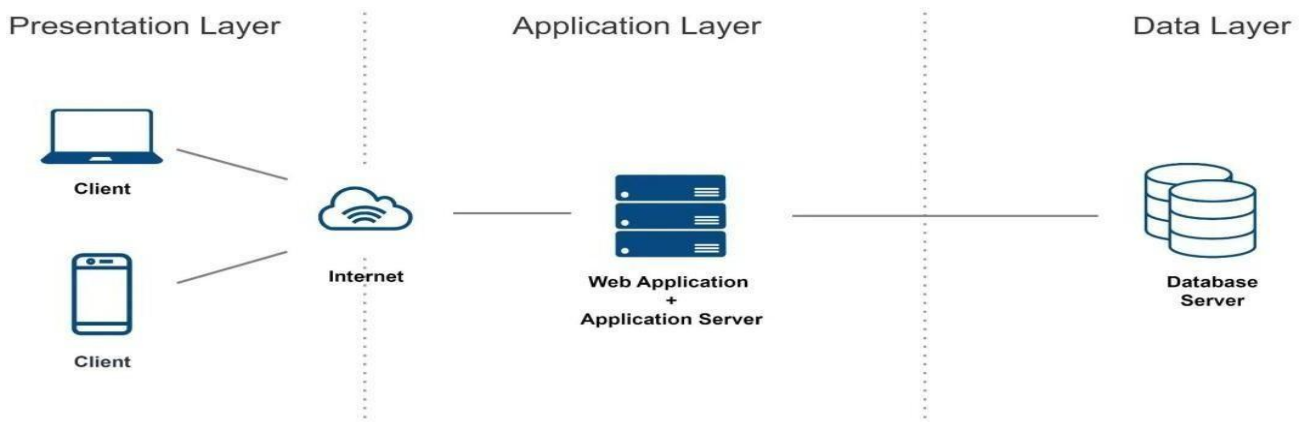


Fig :5.2.1: System Architecture

5.3 DFD Diagram

The DFD diagram below shows the system’s data flow, which is mainly comprised of external entities, processes, data stores, and data flows. The external entities comprise of the Customer, Seller, Admin, and Delivery Person while the processes comprise of Authentication, Product Management, Dynamic Pricing, Order Payment, and Delivery Management. Meanwhile, the data flows in the system include User Input, Product Data, Order Data, Payment Data, and Delivery Information.

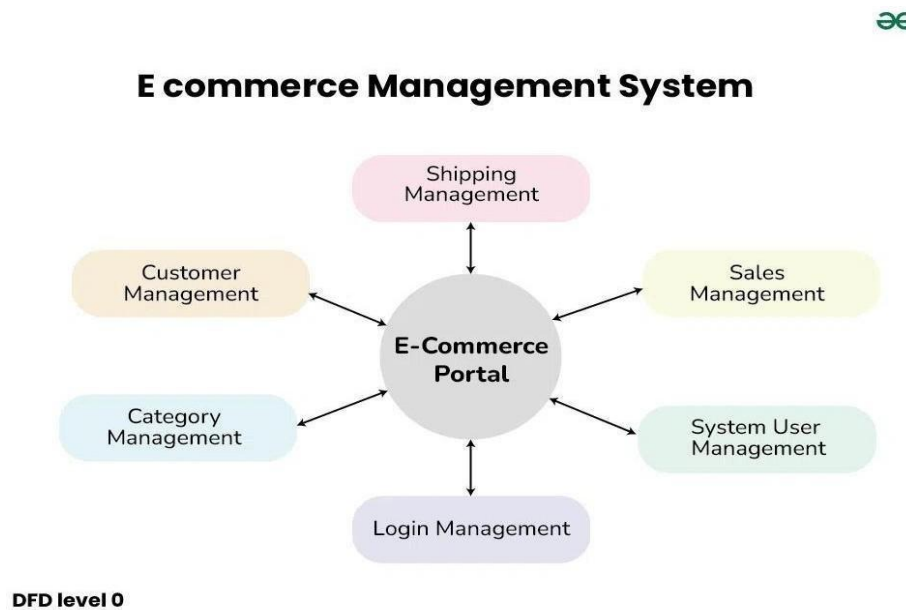


Fig: 5.3.1: DFD Diagram

5.4 UML Diagrams

The UML Diagrams below were developed to depict the system’s design, which includes the Use case diagram, Class diagram, Sequence diagram, Activity diagram, and Deployment diagram.

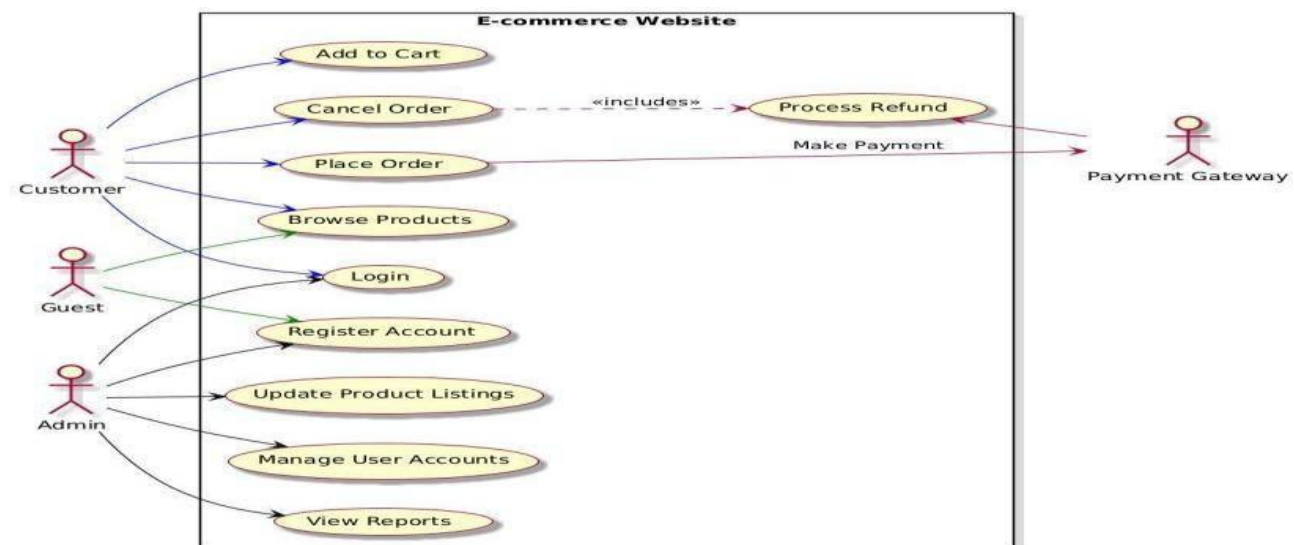


Fig: 5.4.1: UML Diagrams

5.5 Database Design

The system utilizes MySQL as its database to store and manage the system’s data. This is achieved by having a number of tables, which include Users, Products, Categories, Inventory, Cart, Orders, Order_Items, Payments, Delivery, and Dynamic_Price tables. The tables contain fields or columns that hold the attributes of the records stored in the database..

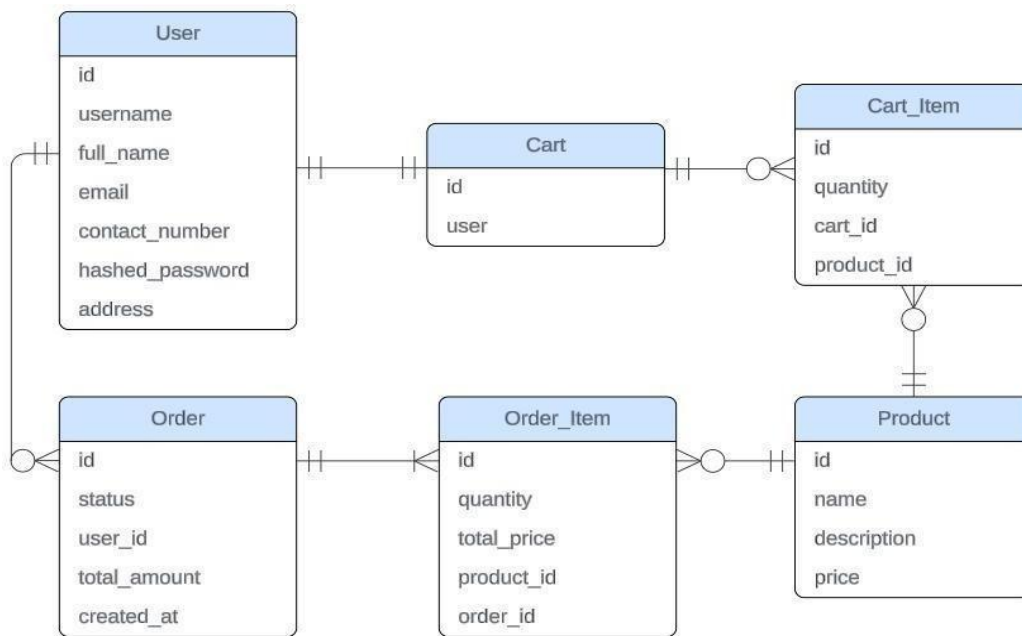


Fig: 5.5.1: Database Design

5.6 Data Dictionary

The Data Dictionary below shows the structure of the database that was utilized in the development of the FashionPriceX – Smart Dynamic Pricing System for Fashion Products to illustrate the essential aspects of a database management system, including tables, columns, and entries.

Table: address				
Description:				
Columns:				
Column	Data type	Attributes	Default	Description
address_id	SMALLINT	PRIMARY, Auto increments, Not null		
address	VARCHAR(50)	Not null		
address2	VARCHAR(50)		NULL	
district	VARCHAR(20)	Not null		
city_id	SMALLINT	Not null		foreign key to column city_id on table city.
postal_code	VARCHAR(10)		NULL	
phone	VARCHAR(20)	Not null		
last_update	TIMESTAMP	Not null	CURRENT_TIMESTAMP	
Indices:				
Name	Columns	Type	Description	
PRIMARY	address_id	PRIMARY		
idx_fk_city_id	city_id	INDEX		

Fig: 5.6.1: Data Dictionary

5.7 Functional Design

FashionPriceX consists of a number of features that are essential for performing its activities efficiently and effectively. Some of these features include User Authentication, Product Management, Dynamic Pricing, Shopping Cart, Payment, Order Management, Delivery Management, and Admin Management.

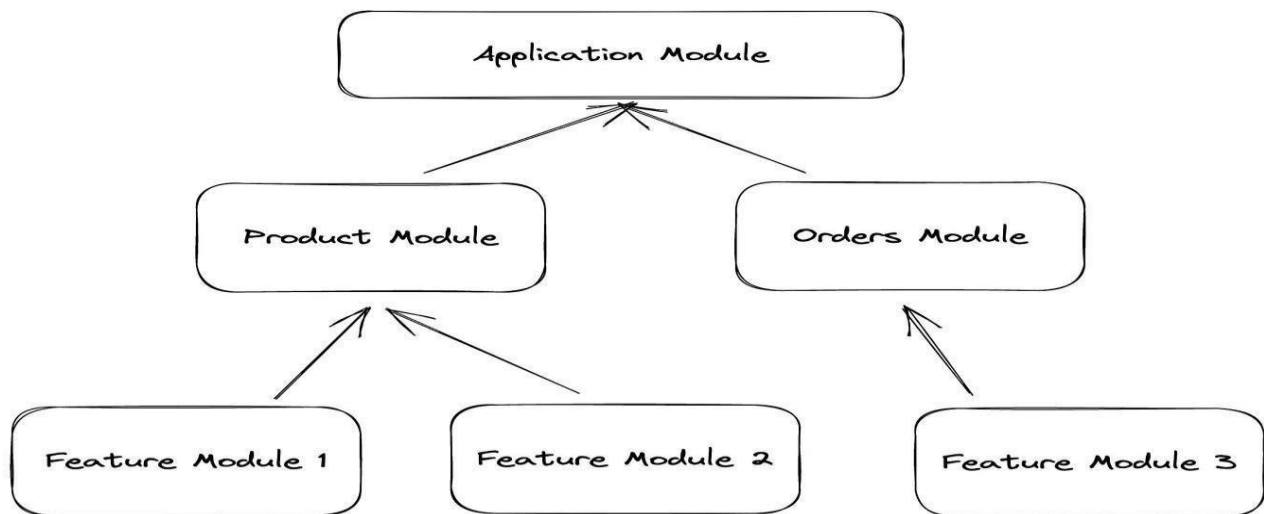


Fig: 5.7.1: Functional Design

5.8 User Interface Design

The user interface has been designed to be intuitive, highly responsive and user-friendly. Separate dashboards have been designed for customers, sellers, administrators and delivery personnel. The user interface has been designed using HTML, CSS, JavaScript, bootstrap and jinja2

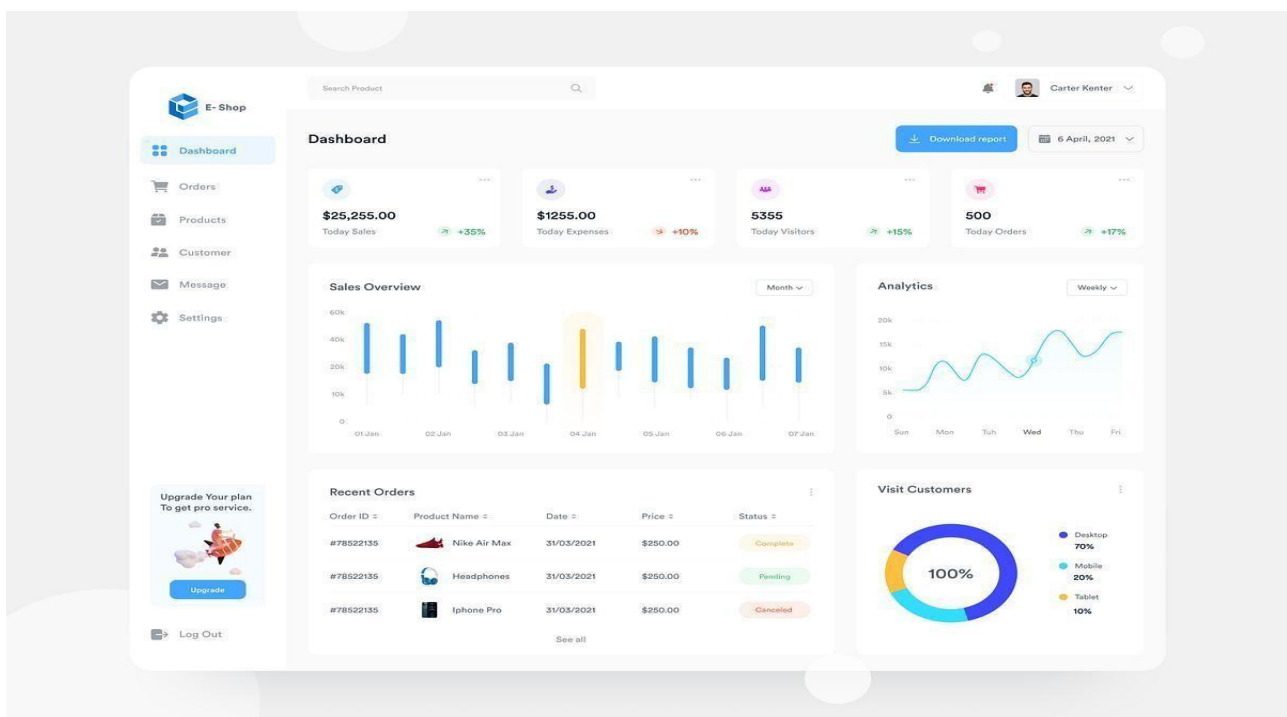


Fig: 5.8.1: User Interface Design

5.9 Sequence Diagram

Sequence Diagram for Online Shopping System

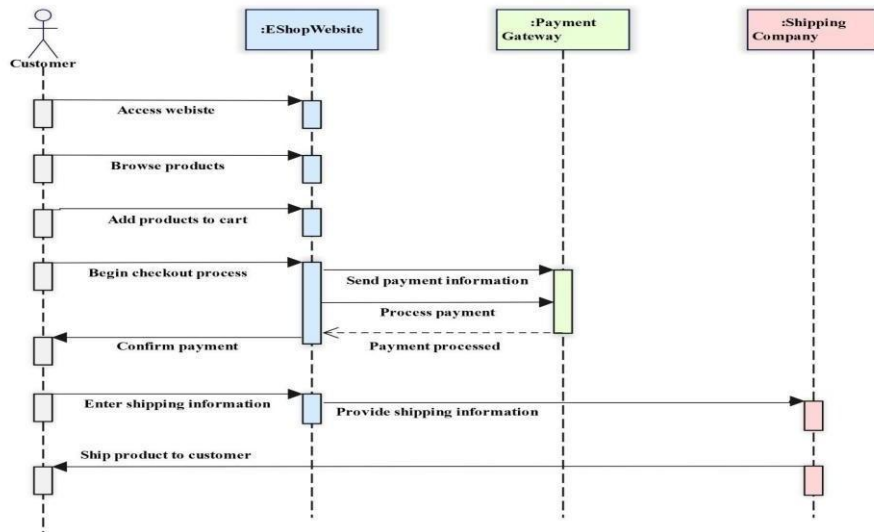


Fig: 5.9.1: Sequence Diagram

5.10 Activity Diagram

The activity diagram below captures the complete process of a customer procuring goods from the company. It begins with the login to the application, followed by product search and selection, adding the chosen product to the cart, proceeding to the checkout section, making payment, receipt of confirmation and delivery.

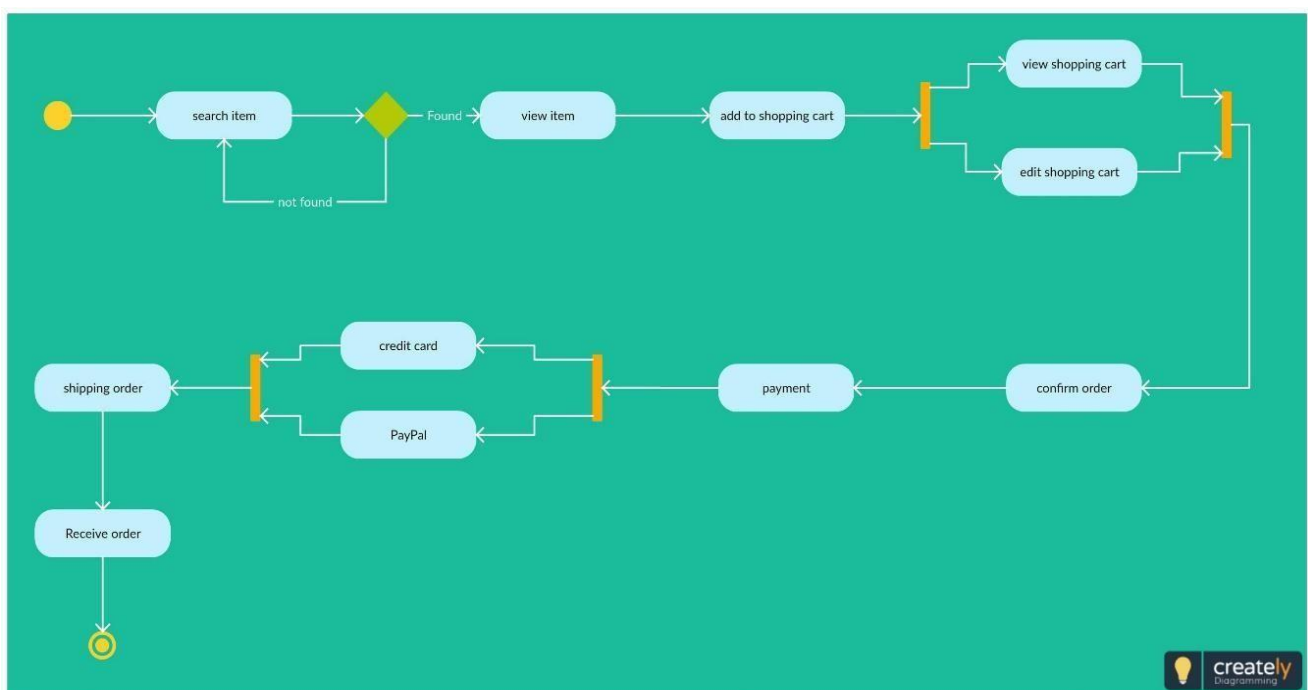


Fig: 5.10.1: Activity Diagram

5.11 Class Diagram

The class diagram below depicts the classes and relationships that have been developed for the project. Some of the many classes that have been developed include User, Customer, Seller, Admin, DeliveryPerson, Product, Cart, Order, Payment, Delivery, Inventory and DynamicPricing.

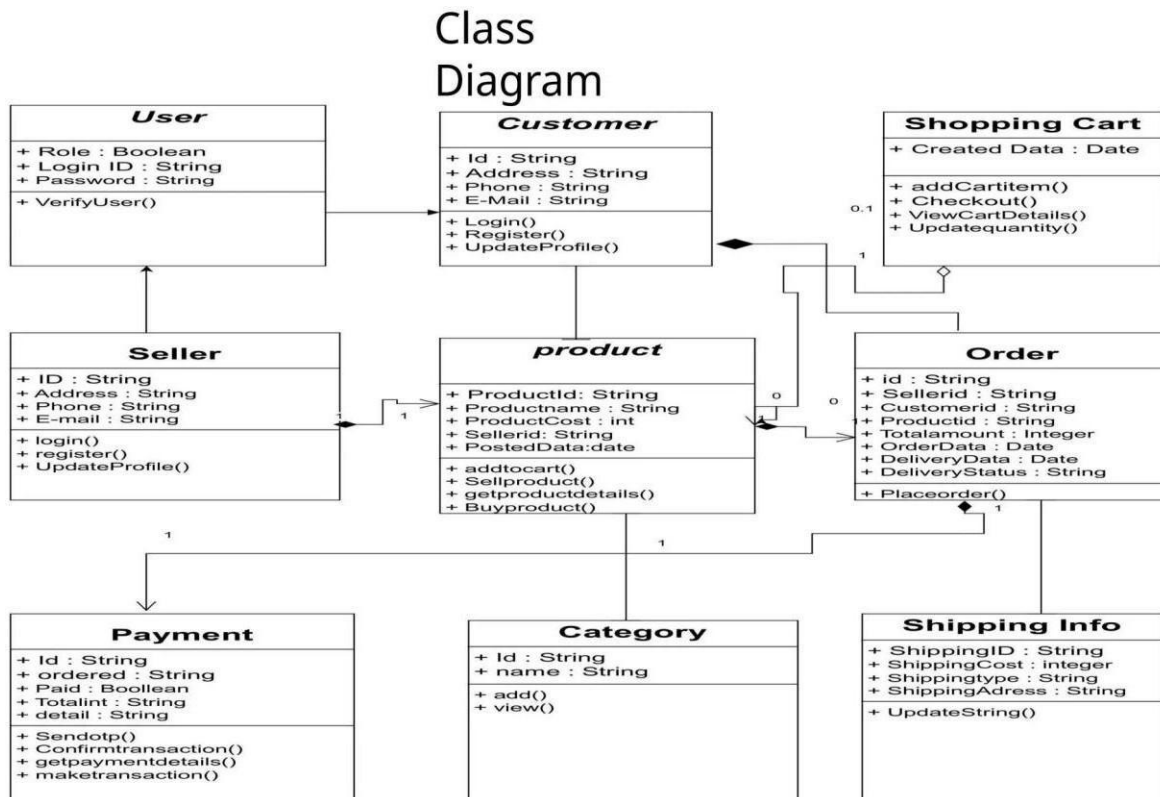


Fig: 5.11.1: Class Diagram

5.12 Deployment Diagram

Finally, the deployment diagram below depicts the overall structure of the system. This entails the application being accessed by the user through a browser, which connects to the flask application server. This further connects to the MySQL database, as well as external payment services.

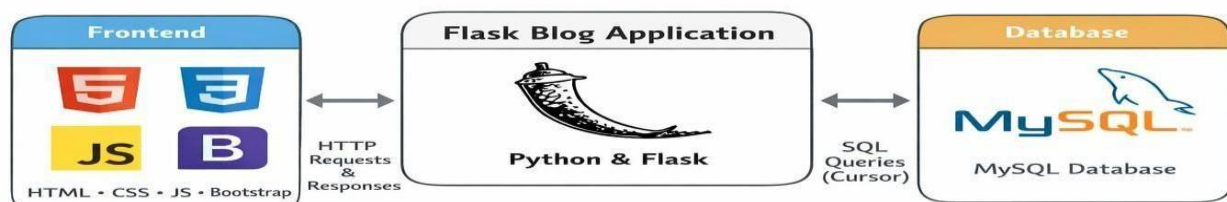


Fig: 5.12.1: Deployment Diagram

5.13 Summary

The paper explored the overall design of the FashionPriceX project. It looked at the system architecture, database, modules, user interface and UML diagrams. It is hoped that the proposed design will serve as a foundation for developing an efficient, integrated, reliable, dynamic, scalable, intelligent fashion E-commerce system.

CHAPTER 6: IMPLEMENTATION

6.1 Introduction

The Implementation chapter contains the description of the development of FashionPriceX – Smart Dynamic Pricing System for Fashion Products. This chapter illustrates how the designed FashionPriceX became a web application using Python, Flask, MySQL, HTML, CSS, JavaScript, and Bootstrap. In addition, the implementation includes user account management, product management, product dynamic pricing, shopping cart, payment and order management, and delivery management processes.

6.2 Module-wise Implementation

The developed FashionPriceX web application consists of several modules, namely Authentication, Seller, Product, Customer, DynamicPricing, Payment and Order, and Delivery. Each module was implemented in Python with an object-oriented programming approach using classes. For instance, the User, Customer, Seller, Admin, Product, Cart, Order, Payment, Delivery, Inventory, and DynamicPricing classes were created to represent the attributes and functions described in the use cases and class diagrams.

6.3 Class-wise Implementation

The FashionPriceX web application was developed using several classes in Python programming language, which primarily follow the principles of object-oriented programming. The key classes, which were created to build the system, include User, Customer, Seller, Admin, Product, Cart, Order, Payment, Delivery, Inventory, and Dynamic Pricing.

6.4 Algorithms / Pseudocode

FashionPriceX utilizes an algorithm that considers the demand and supplies of a product to recommend the appropriate pricing.

In addition to the DynamicPricing algorithm, the application uses various validation algorithms for the login, product details, cart, payment, and order processes.

```

import os
import smtplib
import uuid
from datetime import datetime

from email.message import EmailMessage
from functools import wraps
from flask import Flask, abort, flash, redirect, render_template, request, session, url_for
from flask_mysql import MySQL
from werkzeug.security import check_password_hash, generate_password_hash
from werkzeug.utils import secure_filename

import config

app = Flask(__name__)
app.config["SECRET_KEY"] = config.SECRET_KEY
app.config["MYSQL_HOST"] = config.MYSQL_HOST
app.config["MYSQL_USER"] = config.MYSQL_USER
app.config["MYSQL_PASSWORD"] = config.MYSQL_PASSWORD
app.config["MYSQL_DB"] = config.MYSQL_DB
app.config["MYSQL_CURSORCLASS"] = config.MYSQL_CURSORCLASS
app.config["UPLOAD_FOLDER"] = config.UPLOAD_FOLDER
app.config["MAX_CONTENT_LENGTH"] = config.MAX_CONTENT_LENGTH
mysql = MySQL(app)
os.makedirs(app.config["UPLOAD_FOLDER"], exist_ok=True)

ROLE_ENDPOINTS = {
    "customer": "customer_dashboard",
    "product_builder": "builder_dashboard",
    "delivery_person": "delivery_dashboard",
}

```

6.5 Database Implementation

MySQL was used to implement the database of the FashionPriceX web application. Several tables, including users, products, categories, inventory, cart, orders, order_items, payments, delivery, and dynamic pricing tables, were created in the database. The tables' foreign and primary keys were identified, and the SQL queries were written to add, update, delete, and retrieve data from the database

6.6 User Interface Implementation

The user interface (UI) of the application was developed using HTML, CSS, JavaScript, Bootstrap, and Jinja3. As a result, a responsive UI with customer, seller, admin, and delivery dashboards was created. Different web pages, such as the login page, products listing page, product details page, cart page, check out page, payment page, order tracking pa

6.7 Module Integration

The application modules were integrated using the Flask framework. For instance, the Authentication module was used to control access to the application. In addition, the product module was integrated with the database to display certain information in the web pages. Moreover, the Dynamic Pricing module was connected to the database to update the prices of the products according to the demand and supplies. Similarly, the cart, payment, order, and delivery modules were connected and added to the dashboards. Finally, the modules were tested to ensure that they could operate together efficiently to serve the intended purposes.

6.8 Security and Validation

The validation and security features were added to the FashionPriceX web application to ensure the trustworthiness of the data in the database and the protection of the account details of the users. The features include setting the roles of the users, using strong passwords, encrypting the database, and validating the entries before they are added to the database. For instance, the user details are validated during the registration process and the order processes. Similarly, the database is managed to deny access to unauthorized users. In addition, appropriate algorithms can be used to ensure that the SQL queries are safe and efficient.

6.9 Performance Optimization

The performance of the application was optimized by ensuring that the database queries were properly set, using relational databases, and following the appropriate Flask practices. For instance, the database queries were developed to avoid getting unnecessary information from the database when processing requests. The images and other files were also optimized to increase the speed of the pages and facilitate faster processing of requests by the system. Other optimization approaches include utilizing the Python programming language, Flask framework, and Bootstrap framework to develop a responsive and high-performance application.

6.10 Summary

The optimization and implementation procedure of the project was described in this chapter. Apart from the introduction and summary sections, the chapter consists of ten sections that describe the development and the testing of the proposed modules and classes, the creation of the database using MySQL, the UI development process, the integration of the modules, the security and validation strategies, and finally, the performance optimization and implementation of the web application. The implementation process involved the creation of a web application with dynamic pricing of fashion products.

CHAPTER 7: TESTING

7.1 Introduction

Testing plays a key role in the development process of the FashionPriceX – Smart Dynamic Pricing System for Fashion Products. It is conducted to expose bugs, check the system’s fitness for purpose, and ensure it meets the set objectives. The FashionPriceX prototype will undergo various forms of testing, including functionality, performance, security, usability, and reliability tests before final deployment. This chapter discusses the testing strategy that will be used to evaluate the performance of the FashionPriceX prototype and reports on the test results for each of the proposed test cases. The discussion will focus on testing strategies, test cases, and test results while also covering other aspects related to the testing of the FashionPriceX prototype.

7.2 Testing Strategy

The testing of the FashionPriceX prototype will follow a mix of testing strategies, including unit testing, integration testing, system testing, and user acceptance testing. Additionally, performance and security tests will be conducted to evaluate the performance of the proposed system using set performance benchmarks and ensure the system is secure from possible attacks.

7.3 Unit Testing

The FashionPriceX application is divided into modules that perform specific tasks within the system. Each unit or component will be tested in isolation to establish their strengths and weaknesses before they are merged with other components. The units or modules included in the FashionPriceX system are the user module, product module, dynamic pricing module, cart module, payment module, and delivery module. Each of these modules will be tested individually to detect and remove any bugs that may have been introduced at the unit level.

7.4 Integration Testing

Following successful unit testing, the FashionPriceX application will be tested to establish how well the system’s components work together. The components will be combined and tested as a group to detect integration flaws in the system. Some of the modules within the FashionPriceX system include the product module, dynamic pricing module, cart module, payment module, and delivery module. The modules will be combined and tested to detect design flaws and ensure information is passed from one module to another correctly.

7.5 System Testing

System testing will be used to evaluate the entire FashionPriceX system as a whole. It will be used to determine if the developed system meets the set requirements and objectives before deployment. The entire system will be tested to establish the strength and weaknesses of the system. The key areas that will be tested in the FashionPriceX system include the login system, product catalog, dynamic pricing engine, cart system, payment system, and delivery system.

7.6 User Acceptance Testing

User acceptance testing is a type of testing that determines whether the system meets the user’s requirements and is fit for purpose. In this case, the FashionPriceX application will be tested to determine its fitness for use by customers, sellers, administrators, delivery personnel, and other users of the system. The user acceptance testing will help the users identify some of the flaws and weaknesses that could not be detected at the earlier stages of testing. The feedback from the user acceptance testing will be used to improve the application before its final deployment.

7.7 Test Cases

The following are sample test cases for FashionPriceX:

Test Case	Test Scenario	Expected Result	Status
TC01	Valid user login	User successfully logs in	Pass
TC02	Invalid login details	Error message displayed	Pass
TC03	Add product to cart	Product added successfully	Pass
TC04	Update cart quantity	Quantity updated correctly	Pass
TC05	Place an order	Order successfully created	Pass

TC06 Dynamic price calculation Suitable price displayed Pass

TC07 Successful payment Payment confirmed Pass

TC08 Invalid payment details Payment rejected Pass

TC09 Seller adds product Product displayed in marketplace Pass

TC10 Update delivery status Order status updated Pass

7.7 Test Results

The test results indicate that most functions in the FashionPriceX prototype work as intended and meet the set objectives. The functions that were tested in the FashionPriceX prototype include the user login, addition of products to the cart, updating of the quantity of products in the cart, placing of an order, dynamic pricing engine, processing of payments, and updating of delivery status. The test results further indicate that most bugs and errors that were detected in all the stages of testing were identified and removed. As a result, the system now delivers reliable results when conducting its intended activities.

7.8 Performance Testing

Performance testing will be conducted to measure the system’s responsiveness and stability under various workloads. The test will focus on checking the speed of the application, including the ability to access the database and display web pages. Other aspects that will be tested include searching products, processing of dynamic pricing requests, and order processing. The FashionPriceX system will be tested under normal settings to ensure it meets the performance requirements set for the project. Additionally, the system will be subjected to extreme traffic to determine how well it responds under these conditions.

7.9 Security Testing

FashionPriceX will undergo security tests to detect and eliminate possible security threats and weaknesses. The tests will focus on checking the system’s ability to protect users’ information, including passwords, private information, and payment details. The FashionPriceX system will be tested by trying to access private parts of the system, including the login dashboard, using various means to detect any weaknesses that could expose the system to attacks. Additionally, the system’s ability to filter requests and protect its databases from unauthorized access will also be evaluated to establish its overall strengths and weaknesses. The results of the security testing will then be used to improve the security features of the system.

7.10 Summary

This chapter provided information on the FashionPriceX – Smart Dynamic Pricing System for Fashion Products testing procedures. Different types of tests were described, including unit testing, integration testing, system testing, and user acceptance testing.

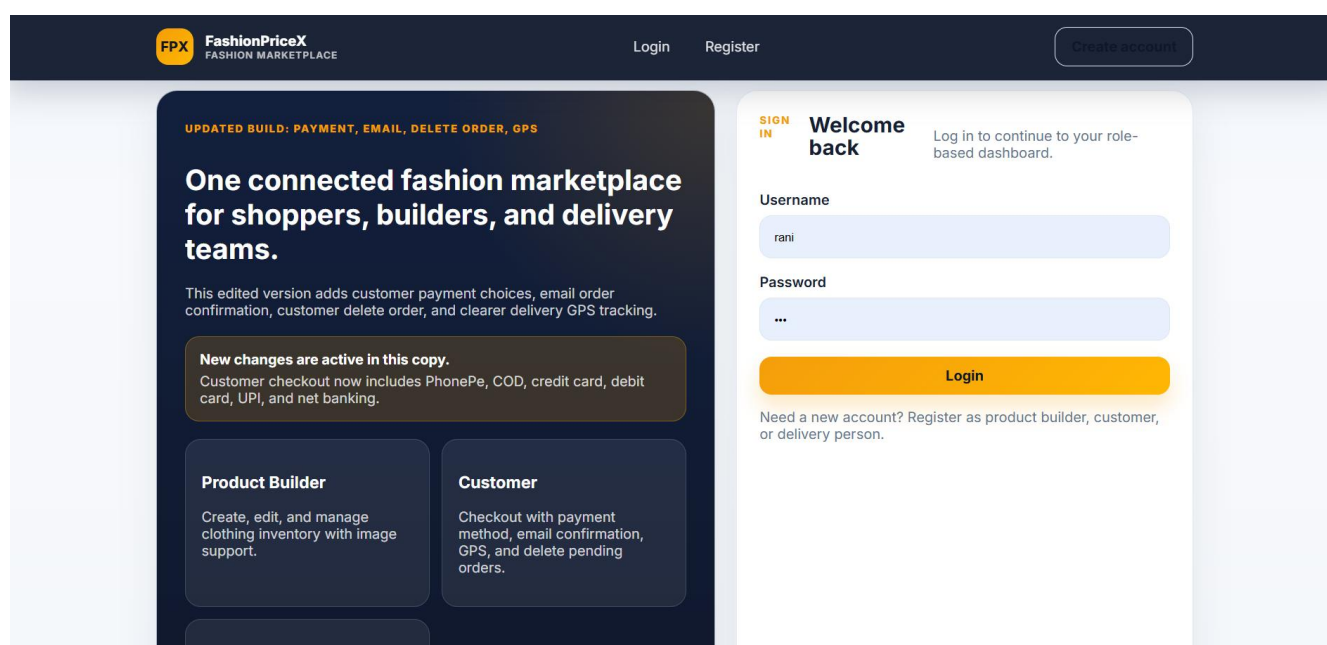
CHAPTER 8: RESULTS AND DISCUSSION

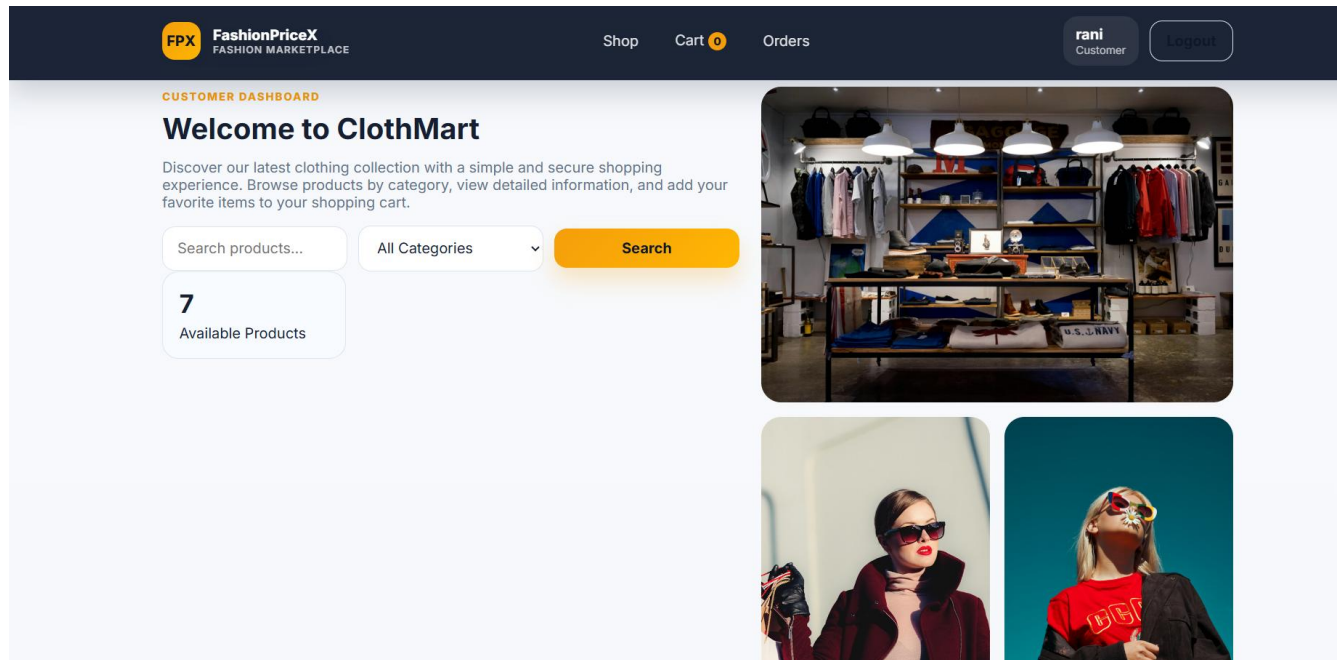
8.1 Experimental Results

The FashionPriceX project has been successfully developed and tested as an online clothing store that is able to offer dynamic pricing to its customers. It has been found that the developed application is able to provide users with the necessary functions of an e-commerce platform, including authentication, product management, shopping cart, payment system, order management, delivery tracking, and dynamic price adjustment. This system has been tested and proven to be functional in its basic aspects, having the ability to offer adjusted prices according to demand and supply.

8.2 Output Screens

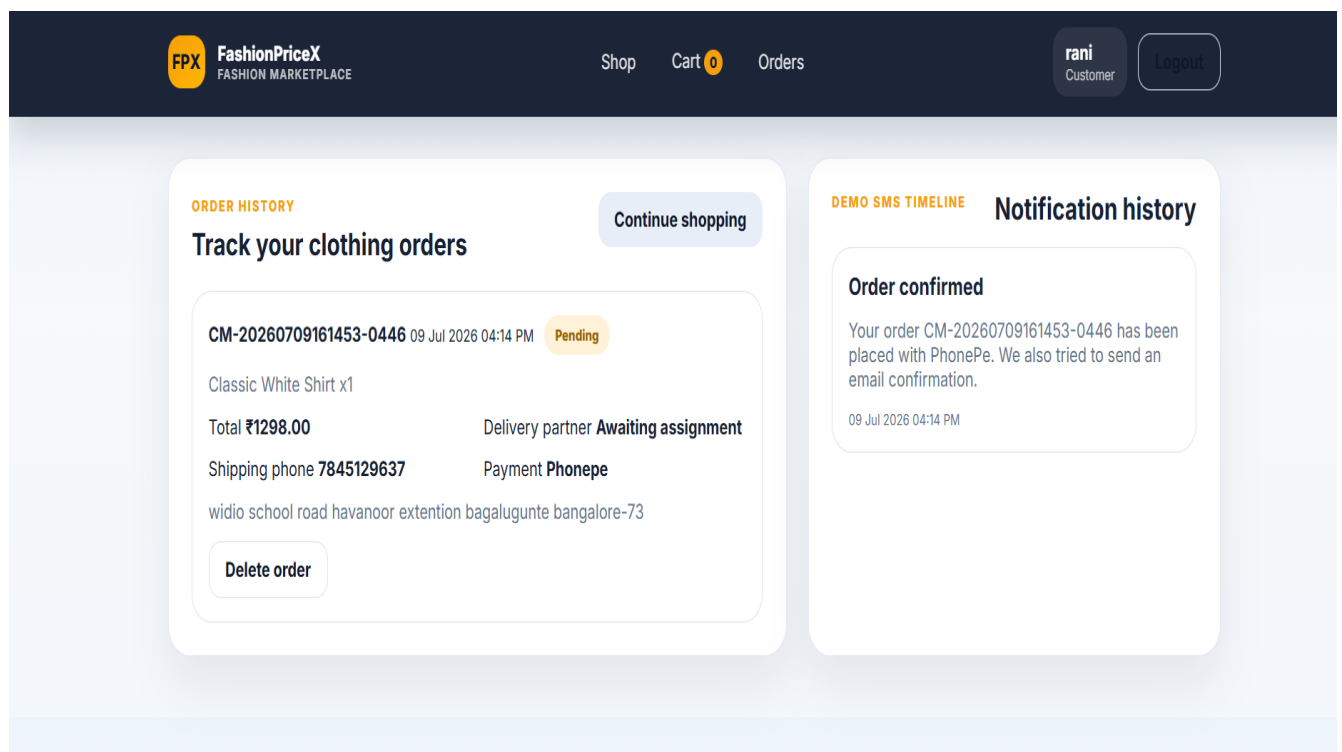
As a web-based e-commerce platform, the developed system offers multiple web interface screens that allow the application’s users to manage their activity within the system and perform the primary functions of an online clothing store. The application’s output screens include the Login Page, Customer Dashboard, Seller Dashboard, Product Listing, Product View, Shopping Cart, Checkout and Payment Page, Order Confirmation Page, Delivery Dashboard, and Admin Page. All of the mentioned above interface screens were created to ensure the primary functions of the developed system.





8.3 Reports

The developed system allows users to operate their tasks within the system and manage their activities accordingly. Being integrated with a database, the system includes the required set of reports that can be used by the system’s users to view, analyze, and manage relevant data. In general, it is possible to state that the developed application allows its users to operate the following reports: product reports, inventory reports, order reports, sales reports, payment reports, delivery reports, and dynamic pricing reports.



8.4 Performance

The developed FashionPriceX web application has been tested, and its performance in terms of core functions was found satisfactory. The main aspects of the performance include the response of the application to user input and the overall speed of operation in terms of loading pages and data. It has been found that in most cases, the web application performs stable and fast, being able to provide the

The screenshot displays the 'CUSTOMER CART' section titled 'Review your clothing selection'. It features a product card for 'shoes' (Footwear) priced at ₹999.00 each, with a quantity of 1. Below the product card are 'Update' and 'Remove' buttons. To the right, the 'CHECKOUT' section is titled 'Shipping and GPS details' and includes a note: 'Delivery visibility uses the latitude and longitude you provide here.' The checkout summary shows a Subtotal of ₹999.00, a Delivery fee of ₹99.00, and a Total of ₹1098.00. Below the summary are input fields for 'Recipient name' (filled with 'rani'), 'Phone number' (filled with '7845129637'), 'Email address for confirmation' (filled with 'shobz123@gmail.com'), and 'Shipping address'.

core functions of an e-commerce platform without significant delays and issues. Notably, the system’s performance is optimized through the proper organization of database interactions and the implementation of modular code.

8.1 Discussion

In general, it has been found that the developed FashionPriceX web application is able to successfully integrate the aspects of an e-commerce platform and dynamic pricing, offering online shoppers the possibility to benefit from both services. The developed application has been tested and found to operate successfully, providing the basic functions of a web-based clothing store and allowing the system’s users to organize their activities according to their needs and demands. Moreover, the dynamic pricing component has been found efficient in terms of adjusting prices according to supply-demand, offering a more reliable alternative to the classical approach of a clothing store. Thus, it may

be concluded that the objectives of the project have been achieved, and the developed system can be considered as a reliable platform that can be improved to address a larger set of challenges and offer additional services, including personalized suggestions, improved demand prediction, and operation within a cloud-based platform.

[Shop](#)
[Cart 1](#)
[Orders](#)

rani Customer

Logout

PAYMENT PAGE

Choose how you want to pay

Back to cart

Select a payment option, then confirm the order.

☒ PhonePe

☐ Cash on Delivery

☐ Credit Card

☐ Debit Card

☐ UPI

☐ Net Banking

Payment details

For this demo, online payment choices are recorded as selected. No card or banking details are stored.

Confirm order

ORDER CONFIRMATION

Review before placing

Subtotal ₹999.00

Delivery fee ₹99.00

Total ₹1098.00

shoes x1

₹999.00

Delivery address

rani · 7845129637

widio school road havanoor extention bagalugunte bangalore-73

CHAPTER 9: CONCLUSION

9.1 Summary of Work

FashionPriceX – Smart Dynamic Pricing System for Fashion Products is a web-based e-commerce platform that was created to offer fashion items for sale and implement dynamic pricing in the process. This system was designed to support the core functions of the e-commerce platform while providing the best experience for the users. The software application highlights such features as authentication, product management, inventory management, shopping cart, payment, order management, delivery tracking, and dynamic pricing of products. The system was built using the Python programming language, the Flask framework, MySQL, HTML, CSS, JavaScript, and Bootstrap. It was established that all the main aspects of the project were properly created and supported the intended purposes and objectives of the system.

9.2 Contributions

The project can be considered as a major contribution to the existing e-commerce platforms as it introduces a dynamic pricing concept into the system while providing the users with extensive features for online shopping. The given project offers multiple benefits to both the sellers and the buyers, thereby supporting the key functions of the e-commerce platform. It allows implementing effective product management and inventory control while introducing smart pricing that follows the demand for goods. In terms of contributions, the project proves the feasibility of using artificial intelligence in online shopping platforms and can become a basis for future e-commerce systems with smart pricing.

9.3 Limitations and Further Research Ideas

Even though the system provides a solid basis for future e-commerce platforms with dynamic pricing, it has several limitations that need to be addressed. First, the project has a significant limitation in terms of providing smart pricing due to potential errors in demand, inventory, and sales data. Therefore, further research should be done to ensure that the system accurately analyzes data and predicts price changes. Moreover, the current version of the software application may be improved by offering more support for external services, including payment options and delivery services, and implementing cloud hosting to ensure continuous operation.

CHAPTER 10: FUTURE SCOPE

10.1 Future Enhancements

The FashionPriceX – Smart Dynamic Pricing System for Fashion Products can be enhanced with a variety of features to ensure increased scalability and performance. Some of the key features that can be added in the future include advanced demand analysis, personalization engine, inventory prediction module, competitive market analysis, and dynamic pricing engine. In addition, the system should be designed to function at scale by deploying it on a cloud platform. Other features that would improve the product offering include mobile application, multiple payment options, sales analytics dashboard, and email/SMS notifications.

10.2 Research Directions

Future research can be conducted to develop novel ideas for enhancing smart dynamic pricing systems. The research can focus on aspects such as Reinforcement Learning, customer behavior, seasonal variations, inventory levels, and market trends. Additional research can be carried out to support features such as personalization, real-time pricing, interpretability, and fairness in the next generation of dynamic pricing products. This research can facilitate the development of FashionPriceX into a leading-edge solution for smart dynamic pricing of fashion products.

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CHAPTER 12: APPENDICES

The appendices comprise supplementary information pertaining to the development, implementation, testing, and operation of FashionPriceX – Smart Dynamic Pricing System for Fashion Products. They offer additional details and information that supplement the project report’s main body.

A. SOURCE CODE

The appendices contain verifiable evidence of the program files that formed the basis of FashionPriceX Development. Some of the files include the Python Flask framework, HTML templates, CSS and JavaScript files, database designs and scripts, configurations, and the dynamic pricing component.

The following are among the files included in the source code:

- Flask application and backend files
- HTML/Jinja2 templates
- CSS and JavaScript Files
- MySQL database scripts including the dynamic pricing component

B. USER MANUAL

The user manual provides guidelines on how to use the developed FashionPriceX system. It describes the roles played by different users in the system, including customers, sellers, delivery personnel, and administrators. Users can browse through categories of products, view products, search, add to cart, checkout, make payments, and track delivery progress.

C. SAMPLE DATA

The appendices also contain sample data used in the development and testing processes to verify the system’s performance. The data includes some of the fields that were used to develop the FashionPriceX application. It comprised the following:

1. Sample Product Data

Product ID	Product Name	Category	Price
P001	Cotton Shirt	Men’s Wear	₹899
P002	Casual Dress	Women’s Wear	₹1,299
P003	Denim Jeans	Men’s Wear	₹1,499



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FashionPriceX

- FashionPriceX
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